

Ejercicios Interpolación Numérica

1. Interpolación de Lagrange (grado 2)

Dado los puntos:

$$(0, 1), (1, 2), (2, 0),$$

Determinar el interpolador $P(x)$ mediante la fórmula y calcular $P(1.5)$.

Sol:

Las bases de Lagrange son:

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$

$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2)$$

$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

El polinomio interpolador es:

$$\begin{aligned} P(x) &= 1 \cdot L_0(x) + 2 \cdot L_1(x) + 0 \cdot L_2(x) \\ &= \frac{(x-1)(x-2)}{2} - 2x(x-2) \end{aligned}$$

Simplificando:

$$P(x) = \frac{x^2 - 3x + 2}{2} - 2x^2 + 4x = \frac{-3x^2 + 5x + 2}{2}$$

$$P(1.5) = \frac{-3(1.5)^2 + 5(1.5) + 2}{2} = \frac{-6.75 + 7.5 + 2}{2} = \frac{2.75}{2} = 1.375$$

2. Forma de Newton (diferencias divididas)

Interpolan las puntos:

$$(0, 1), (1, 3), (2, 2), (3, 5),$$

mediante el polinomio de Newton en forma de diferencias divididas, y calcular el valor aproximado en $x = 1.5$.

Solución:

x	$f(x)$	$f(x, x+1)$	$f(x, x+1, x+2)$	$f(x, x+1, x+2, x+3)$
0	1			
1	3	2		
2	2	-1	$-\frac{3}{2}$	
3	5	3	2	

Valores extraídos:

$$f(x_0) = 1 \quad f(x_1) = 3 \quad f(x_2) = 2 \quad f(x_3) = 5$$

$$f'(x_0, x_1) = \frac{3-1}{1-0} = 2, \quad f'(x_1, x_2) = \frac{2-3}{2-1} = -1, \quad f'(x_2, x_3) = \frac{5-2}{3-2} = 3$$

$$f''(x_0, x_1, x_2) = \frac{-1-2}{2-0} = -\frac{3}{2}, \quad f''(x_1, x_2, x_3) = \frac{3-(-1)}{3-1} = 2$$

$$f'''(x_0, x_1, x_2, x_3) = \frac{2-(-\frac{3}{2})}{3-0} = \frac{7/2}{3} = \frac{7}{6}$$

2.º forma de Newton:

$$\begin{aligned} P(x) &= f(x_0) + f'(x_0, x_1)(x-x_0) + f''(x_0, x_1, x_2)(x-x_0)(x-x_1) + \\ &\quad + f'''(x_0, x_1, x_2, x_3)(x-x_0)(x-x_1)(x-x_2) \\ &= 1 + 2x - \frac{3}{2}x(x-1) + \frac{7}{6}x(x-1)(x-2) \end{aligned}$$

Evaluamos en $x = 1.5$:

$$\begin{aligned} P(1.5) &= 1 + 2(1.5) - 3 \frac{(1.5)(0.5)}{2} + 7 \frac{(1.5)(0.5)(-0.5)}{6} \\ &= 1 + 3 - \frac{3 \cdot 0.75}{2} + \frac{7}{6} \cdot (-0.375) \\ &= 4 - 1.125 - 0.4375 = 2.4375 = \frac{39}{16} \end{aligned}$$

E- Spline cúbico natural (3 puntos)

Dados los puntos:

$$(0, 0) \quad (1, 1) \quad (2, 0)$$

Construir el spline cúbico natural $S(x)$ y evaluar $S(0.5)$ y $S(1.5)$

Solución:

$$h_0 M_0 + 2(h_0 + h_1) M_1 + h_1 M_2 = 6 \left(\frac{f_2 - f_1}{h_1} - \frac{f_1 - f_0}{h_0} \right)$$

Sustituyendo $h_0 = h_1 = 1$, $f_0 = 0$, $f_1 = 1$, $f_2 = 0$ y $M_0 = M_2 = 0$

$$4M_1 = 6((-1)(1)) = 6(-2) = -12 \Rightarrow M_1 = -3$$

Por tanto $M_0 = 0$, $M_1 = -3$, $M_2 = 0$

La fórmula en el subintervalo (x_i, x_{i+1}) con $h_i = 1$ es

$$S_i(x) = \frac{M_i}{6} (x_{i+1} - x)^3 + \frac{M_{i+1}}{6} (x - x_i)^3 + \left(\frac{f_i - M_i}{6} \right) \frac{x_{i+1} - x}{h_i} + \left(\frac{f_{i+1} - M_{i+1}}{6} \right) \frac{x - x_i}{h_i}$$

Intervalo $[0, 1]$ con $M_0 = 0$, $M_1 = -3$, $f_0 = 0$, $f_1 = 1$

$$S(x) = -\frac{1}{2}x^3 + \frac{3}{2}x$$

Intervall $[1,2]$ mit $M_1 = -3, M_2 = 0, F_1 = 1, F_2 = 0$

$$S(x) = -\frac{1}{2}(2-x)^3 + \frac{3}{2}(2-x)$$

Berechnen $S(0.5)$ (Parameter in $[0,1]$)

$$S(0.5) = -\frac{1}{2}(0.5)^3 + \frac{3}{2}(0.5) = -0.0625 + 0.75 = \frac{11}{16}$$

Berechnen $S(1.5)$ (Parameter $[1,2]$, durch $2-x=0.5$)

$$S(1.5) = -\frac{1}{2}(0.5)^3 + \frac{3}{2}(0.5) = -0.0625 + 0.75 = 0.6875$$

$$= \frac{11}{16}$$

```

1 # Ejercicio 1: Interpolación de Lagrange (grado 2)
2
3 # Datos
4 x <- c(0, 1, 2)
5 y <- c(1, 2, 0)
6
7 # Función de Lagrange general
8 lagrange <- function(x, y, x_eval) {
9   n <- length(x)
10  p <- 0
11  for (i in 1:n) {
12    L <- 1
13    for (j in 1:n) {
14      if (j != i) {
15        L <- L * (x_eval - x[j]) / (x[i] - x[j])
16      }
17    }
18    p <- p + y[i] * L
19  }
20  return(p)
21 }
22
23 x_eval <- 1.5
24 p_1_5 <- lagrange(x, y, x_eval)
25 p_1_5
26 |

```

```

1  # Ejercicio 2: Interpolación de Newton (diferencias divididas)
2
3  # Datos
4  x <- c(0, 1, 2, 3)
5  y <- c(1, 3, 2, 5)
6
7  n <- length(x)
8  dd <- matrix(0, n, n)
9  dd[,1] <- y
10
11 for (j in 2:n) {
12   for (i in 1:(n - j + 1)) {
13     dd[i,j] <- (dd[i+1,j-1] - dd[i,j-1]) / (x[i+j-1] - x[i])
14   }
15 }
16
17 coef <- dd[1,]
18
19 newton_eval <- function(x, coef, x_eval) {
20   n <- length(coef)
21   p <- coef[n]
22   for (k in (n-1):1) {
23     p <- coef[k] + (x_eval - x[k]) * p
24   }
25   return(p)
26 }
27
28 x_eval <- 1.5
29 p_1_5 <- newton_eval(x, coef, x_eval)
30 p_1_5
31

```

```
# Ejercicio 3: Spline cúbico natural (3 puntos)
```

```
x <- c(0, 1, 2)
```

```
y <- c(0, 1, 0)
```

```
spline_nat <- splinefun(x, y, method = "natural")
```

```
s_0_5 <- spline_nat(0.5)
```

```
s_1_5 <- spline_nat(1.5)
```

```
s_0_5
```

```
s_1_5
```